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ORIGINAL ARTICLE

Assessment of gridded precipitation and air temperature products for the State of Acre, southwestern Amazonia, Brazil

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Abstract This paper evaluated the precision and accuracy of two gridded monthly precipitation products (P-GPCC/V6 from the Global Precipitation Climatology Centre and P-UDEL/V3.01 from the University of Delaware—UDEL) and two monthly air temperature products (T-GHCN/V2 from the Global Historical Climatology Network and T-UDEL/V3.01 from UDEL) for the State of Acre, in southwestern Amazonia, northern Brazil. As reference, we used monthly climate series of these elements from January 1971 to December 2000, recorded at five conventional weather stations (CWS) of the Instituto Nacional de Meteorologia (INMET). Data were compared by linear regression; the coefficient of determination (r^2); Willmott's index of agreement (d); Camargo's performance index (c); the root-mean-square error (RMSE); mean systematic error (MSE); and unsystematic error (MSE_u). Moreover, a trend analysis of the products at a seasonal and annual scale was performed. Regarding rainfall, it was found that GPCC as well as UDEL represents the mean and variability of rainfall throughout the series, but the performance of the product of GPCC was more homogenous and generally applicable for all stations. In relation to the air temperature standards, the precision of GHCN and UDEL was low, but the accuracy was moderate, according to statistical methods.

Keywords Gridded products · Precision · Accuracy · Climate models · Statistical methods · Meteorological systems

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Introduction

The Amazon is a key region for climate control on a local, regional and global scale. This is related to the high amount of solar energy in the region and water availability, resulting from convective processes and evapotranspiration (ET). Nevertheless, the rainfall regimes in the region are not homogeneous, but highly variable in spatial (regional and large scale) and temporal (intra-annual and interannual) terms, due to the influence of different meteorological systems at the local, meso- and large scale and climate variability modes (Rocha 2001; Duarte 2006; Reboita et al. 2010). Among the climate variability modes conditioning rainfall and air temperature patterns of the region of this study are the El Niño-Southern Oscillation (ENSO), the Atlantic dipole and the Pacific Decadal Oscillation (PDO) (Marengo 2005).

In Acre, a state in the southwest of the Brazilian Amazon, the performance of the meteorological systems, climate variability and associated mechanisms is not comprehensively understood, partly because scientific studies for this region focus only on static climate descriptions based on rainfall and air temperature (Duarte 2006). Due to the importance of this region in the regional and global climate, studies are needed to understand the intra-annual (monthly and seasonal) variation of rainfall and air temperature in the region, and how the interactions

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between meteorological systems and physiographic factors condition this variation (Almeida et al. 2015). In addition, it is important to understand how the climate variability modes (ENSO, dipole and PDO) influence the regional meteorological systems and consequently, the interannual, decadal and multidecadal rainfall and air temperature patterns (Almeida et al. 2016).

Studies addressing the high-frequency (monthly and seasonal) and low-frequency (interannual, decadal and multidecadal) climatic variability and on a regional scale require long (>30 years) continuous, high-quality climate series, regularly distributed in space and with high resolution (<0.5°). Anyway, due to the geographical and economic limitations in the region, spatially regularly distributed time series covering 30 years or more are scarce, hampering researches in different areas of the region. This is a recurring problem in many regions of the planet, mainly in most of South America, Africa and part of Asia (GPCC 2014; Prakash et al. 2015).

Although reanalysis data comprise long-term series (Reanalysis 1, from 1948 to the present) in grids, which consider the whole planet, the low spatial resolution of the grid (2.5° × 2.5°) restricts regional studies (Kalnay et al. 1996; Kanamitsu et al. 2002). An exception is the regional reanalysis for North America (NA), available with a spatial resolution of 0.3° × 0.3° (Mesinger et al. 2006). The numerical climate models are usually Atmospheric General Circulation Models (MCGA), which also hamper regional climate studies due to the low resolution of the simulations and the need for a robust computer structure for long-term simulations. In both cases, for regional climate studies downscaling techniques should be applied to obtain products with a higher resolution, which is time-consuming and requires computational power, mainly for dynamic downscaling. The rainfall products obtained by remote orbital sensors such as CMORPH, TRMM-3B42 and TRMM-3B43 have a high resolution (0.25° × 0.25° or 0.5° × 0.5°), provided by multiple estimators underlying the elaboration of the products of the Tropical Rainfall Measuring Mission (TRMM) satellite, and good accuracy and precision of these products for several relief conditions (Feidas 2010; Dinku et al. 2008). Nevertheless, they have no long-term data series (>30 years), which restricts studies on climatic variability and extreme events, e.g., on droughts, floods, desertification, and on the forest fires/arson common in the Amazon region (Almeida et al. 2016).

The grid data, obtained by techniques of spatial interpolation of rain and air temperature data, measured in networks of surface meteorological stations, include long-term data series. Some of these cover over 100 years, as in the case of grid data elaborated at the University of Delaware (Willmott and Matsuura 2001) and in various spatial resolutions, from 2.5° × 2.5° to high (0.5°) resolution (GPCC 2014; Willmott and Matsuura 2001). The temporal characteristics and high spatial resolution (0.5°) of these grid data indicate them for applications in regional climate studies, for example addressing dry-season climatic variability (GPCC 2014).

However, historical gridded data obtained with different products may contain several errors, and a comparison with data from in situ observations can reveal several errors, especially for remote areas, where the reliability of the observed data is particularly questionable. Therefore, the different methodologies applied in the spatial interpolation processes of gridded data, as well as terrain characteristics and the number of stations used to generate the grids, can generate interpolated products with disparate data, affecting the reliability of estimates (Sun et al. 2014).

Thus, given the need to obtain reliable climate data with spatial regularity, high resolution and long-term data series for regional climate studies (e.g., on drought, floods and desertification), the data precision and accuracy were evaluated of the two gridded monthly precipitation products (P-GPCC/V6 from the Global Precipitation Climatology Centre and P-UDEL/V3.01 from the University of Delaware {UDEL}) and the two monthly air temperature products (T-GHCN/V2 from the Global Historical Climatology Network and T-UDEL/V3.01 from UDEL).

Materials and methods

Description of gridded products

The GPCC product (Total Full V6) used in this study provides monthly precipitation data at the global level, in a grid with a spatial resolution of 0.5° × 0.5°, from 1948 to 2010. To generate this product, synoptic meteorological data were used, accessible by the Global Telecommunications System (GTS); monthly weather bulletins; weather station data from different countries; data collections of some international regional projects; as well as from the collection of global data of the Climatic Research Unit (UK) and the United Nations Food and Agriculture Organization (FAO) (Rudolf and Schneider 2005).

The quality of weather station data is rigorously controlled by preprocessing of the data source; time-series analyses that identify spurious changes in mean and variance; and spatial comparisons to check the accuracy of the climatological mean and the seasonal cycle in relation to neighboring points, to identify outliers. After these procedures, potential errors are corrected, if possible; otherwise, these records are discarded and included in the final composition (Schneider et al. 2014).

Thereafter, the rainfall data are interpolated by the method SphereMap (Willmot et al. 1985), and the errors associated with the interpolation process are estimated. The method SphereMap is a spherical adaptation of the Shepard weighting system (Shepard 1968), which considers: (a) the distances of the stations to the grid point (for a limited number of nearest stations), (b) the directional distribution of stations in relation to the grid point (to avoid excess weight of stations in clusters) and (c) the gradients of the data field at the grid point.

To generate the gridded product, the rainfall anomalies of the stations are interpolated and then superimposed on the total absolute precipitation (Beck et al. 2005). A detailed description of the methodology used to generate the GPCC precipitation product can be found in Rudolf and Schneider (2005), Becker et al. (2013) and Schneider et al. (2014).

The GHCN product (V2) used in this study provides monthly air temperature data at the global level, at a grid resolution of 0.5 × 0.5°, from 1948 to 2010. Data from 31 sources from across the world are compiled, and the quality is rigorously controlled, consisting of preprocessing and control of the source data. Homogeneity tests are also performed, aimed at minimizing the effect of non-climatic factors on the time series, such as changes of land use and soil cover (Lawrimore et al. 2011).

In this product, air temperature anomalies are used to normalize the data so that they can be compared and combined to represent the temperature patterns more accurately in relation to what is normal for different locations within a region, particularly in areas with varying altitude and/or sparse data (Fan and van Den Dool 2008).

For this calculation, the Climate Anomaly Method (CAM) (Ropelewski et al. 1984) or the First Difference Method (FDM) is used (Peterson et al. 1998). The CAM involves the calculation of temperature anomalies (from the 30-year climatological mean of 1961–1990 or 1971–2000) for each month and year of the registration period of a weather station. The FDM, on the other hand, is used to calculate anomalies of stations with a smaller data set, with at least two years of consecutive data. This method is based exclusively on the calculation of the air temperature difference between one year and the next.

After these calculations, the anomaly fields are combined with those of air temperature to provide final grid values by the method of Cressman (1958) for interpolation. A detailed description of the methodology used in the gridded air temperature product of GHCN was presented by Fan and Van den Dool (2008) and Lawrimore et al. (2011).

Apart from the products described above, two other databases generated by the University of Delaware were used for comparison, one of which is a product of precipitation (V3.01) and the other of air temperature (V3.01).

The precipitation grid of UDEL compiles data from various sources: the Global Historical Climatology Network (GHCN2), Canada's Atmospheric and Environmental Service; the hydrometeorological Institute of Saint Petersburg—Russia; Greenland Climate Network; records from automatic weather stations of the Greenland Project, the National Center for Atmospheric Research (NCAR), daily data from India; rainfall data files of the African continent; monthly rainfall data of the South American continent; Global Surface Summary of the Day (GSOD) and of Legates and Willmott (1990). After a quality control and data blending, the absolute rainfall values were interpolated by the spherical method of Shepard (Shepard 1968) in a 0.5° x 0.5° grid (Matsuura and Willmott 2009).

The UDEL air temperature product is based on GHCN version 2 records of Legates and Willmott (Legates and Willmott 1990a, b). The gridded product is also available at a resolution of 0.5° × 0.5°, but this is generated by the method climatologically aided interpolation (CAI) (Willmott and Robeson 1995), adjusted by the digital elevation model (DEM).

The CAI uses climatological data of high spatial resolution to obtain a monthly air temperature difference at each station. These differences between stations are interpolated by the Shepard algorithm. Finally, each gridded monthly difference is added back to the corresponding field of monthly climate to obtain the final temperature value (Willmott and Robeson 1995).

It is noteworthy that the interpolation process generates a product based on absolute air temperature data instead of anomaly fields. After interpolating both products, errors are estimated by cross-validation.

Detailed descriptions of precipitation and air temperature products of UDEL were provided by Legates and Willmott (1990a, b), Willmott and Matsuura (1995), and Willmott and Robeson (1995).

Data validation

The precipitation and air temperature data of the products were validated for the period from January 1971 to December 2000, using climate series of five conventional weather stations (CWS) of INMET. Since there are only three stations in the State of Acre, data from two stations in the State of Amazonas were included.

A point grid with the GPCC data of precipitation, GHCN air temperature and UDEL precipitation and air temperature data was superimposed on the location of the stations (Fig. 1). For validation, data from the grid points closest to the stations were used (see Table 1).

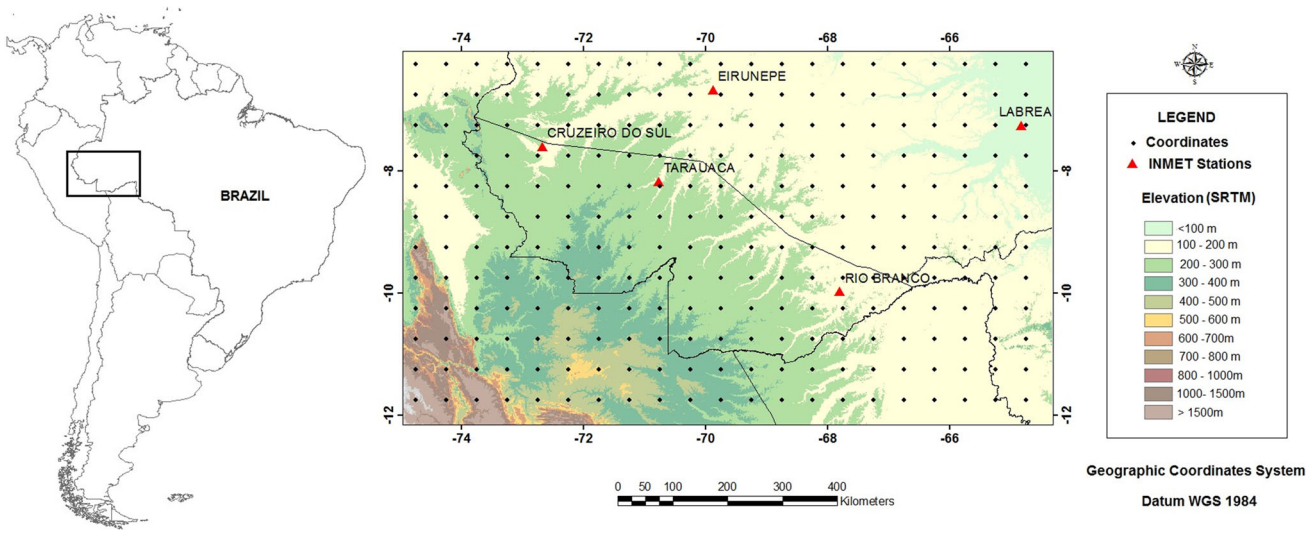


Fig. 1 Location of conventional weather stations of INMET and center dots grid with the data of precipitation and air temperature used in the analyses

Table 1 Location of INMET stations and air temperature grade point and precipitation of products used in the analyzes

Stations (INMET)	Altitude (m)	Coordinate stations		Grid points GPCC, GHCN and UDEL	
		Lat (°)	Long (°)	Lat (°)	Long (°)
Cruzeiro Sul—AC	170	–7.6	–72.67	–7.75	–72.75
Eirunepé—AM	104	–6.67	–69.87	–6.75	–69.75
Lábrea—AM	61	–7.25	–64.83	–7.25	–64.75
Rio Branco—AC	160	–9.97	–67.8	–9.75	–67.75
Tarauacá—AC	190	–8.17	–70.77	–8.25	–70.75

The data were validated by comparing the monthly precipitation and air temperature values estimated by the gridded products with the observed CWS data of INMET. The trend of rainfall at a seasonal and interannual scale was also evaluated. Regarding air temperature, the trend of this variable was only analyzed on an annual scale, since the amplitude of mean temperature changes over the months is low in this study area near the Equator.

Aside from the analysis of seasonal and interannual trends, the data were also compared by the following methods or statistical indices: simple linear regression between the estimated data (Y) by the gridded data (GPCC, GHCN or UDEL) and observed data (X) at the weather stations; the coefficient of determination (r^2); the Willmott index of agreement (d); the Camargo performance index (c); root-mean-square error (RMSE); mean systematic error (MSE_s); and the non-systematic error (MSE_u). The r^2 quantifies the precision of the method, that is, the degree of dispersion of the points from the mean indicating the random error.

The Willmott index of agreement (d) determines the accuracy of the method, that is, it indicates the degree of deviation from the estimated in relation to the observed values (Willmott 1981). This index ranges from 0 for no agreement to 1 for perfect agreement. The agreement index is calculated by the following equation:

$$d = 1 - \left[\sqrt{\frac{\sum_{i=1}^{N} \left(P_i - O_i\right)^2}{\sum_{i=1}^{N} \left(|P_i - O| - |O_i - O|\right)^2}} \right] \quad (1)$$

where N = number of observations; P_i = values estimated by GPCC, GHCN or UDEL; O_i = values observed at INMET stations; O = mean of observed values.

The index "c" developed by Camargo and Sentelhas (1997) evaluates the overall performance of the method, summarizing the results of the indices "d" and "r," according to the equation:

$$c = r * d \quad (2)$$

where c = Camargo index; r = correlation coefficient; d = Willmott index.

The criterion by which the performance of estimated data according to the "c" index was interpreted is listed in Table 2.

Table 2 Performance criteria interpretation by the index "c" proposed by Camargo and Sentelhas (1997)

Value	Performance
>0.85	Great
0.76 a 0.85	Very good
0.66 a 0.75	Good
0.61 a 0.65	Mean
0.51 a 0.60	Poor
0.41 a 0.50	Bad
<0.4	Very bad

The root-mean-square error (RMSE) provides information on data dispersion, i.e., the scattering degree obtained in comparison with the estimated values, where ideally, the RMSE would be equal to 0. It is estimated by the following equation:

$$\mathrm{RMSE} = \sqrt{N^{-1}\sum_{i=1}^{N}(P_i - O_i)^2} \quad (3)$$

Nevertheless, the RMSE does not indicate whether errors are systematic or non-systematic (random). According to Willmott (1981), it is therefore important to define each error type separately:

(a) systematic error ($\mathrm{MSE_s}$)—the pattern of an instrument or model to over- or underestimate data systematically, affecting the accuracy of measurement. This error represents the sum of sources of systematic errors from precipitation and air temperature estimates based on the regression equations (Y = a + b X):

$$\mathrm{MSE_s} = \mathrm{MSE_a} + \mathrm{MSE_p} + \mathrm{MSE_i} \quad (4)$$

where

- ($\mathrm{MSE_a}$)—additive systematic error—results from under- or overestimation compared to the values observed at stations in relation to the constant term (a) of estimates of GPCC, GHCN or UDEL, defined as:

$$\mathrm{MSE_a} = \mathrm{a}^2 \quad (5)$$

- ($\mathrm{MSE_p}$) proportional systematic error—results from under- or overestimation compared to the observed values at weather stations in relation to the proportional term (b) of estimates of GPCC, GHCN or UDEL, defined as:

$$\mathrm{MSE_p} = (b-1)^2\left[\frac{\sum_{i=1}^{N} O_i^2}{N}\right] \quad (6)$$

where N = number of observations; O_i = values observed at INMET stations.

- ($\mathrm{MSE_i}$) interdependent systematic error—the degree of interdependence between the additive (a) and proportional error is given in terms of the covariance between the error due to the additive term (a) and the proportional error (b-1):

$$\mathrm{MSE_i} = 2a\,(b-1)\bar{O} \quad (7)$$

where $\bar{O}$ is the mean of the observed values.

(b) non-systematic error ($\mathrm{MSE_u}$)—the random error results from variations in measurements that do not follow a fixed trend, but that can be analyzed by their dispersion around the mean, which affects the measurement precision. The non-systematic (random) error is calculated by:

$$\mathrm{MSE_u} = \mathrm{MSE} - \mathrm{MSE_s} \quad (8)$$

where $\mathrm{MSE_s}$ = systematic error; MSE = total mean error;

The MSE is given by:

$$\mathrm{MSE} = \frac{\sum_{i=1}^{N}\left(P_i - O_i\right)^2}{N} \quad (9)$$

where N = number of observations; P_i = values estimated by GPCC, GHCN or UDEL; O_i = values recorded at INMET stations.

According to Andrade-Júnior et al. (2003), MSE, $\mathrm{MSE_e}$ and $\mathrm{MSE_u}$ should be interpreted based on their square roots (RMSE, $\mathrm{RMSE_e}$ and $\mathrm{RMSE_u}$), since these are defined in the same measurement units of O and P (mm or °C). Therefore, after calculating the error types, their roots were extracted, and only the former were used to interpret the results.

Results and discussion

Comparison of precipitation products

The interannual analysis of the climate data series (Fig. 2) indicated that the GPCC and UDEL products satisfactorily represented the rainfall pattern observed at the INMET stations in most years, since the Pearson's correlation coefficient exceeded 0.72 for both products (Table 3).

Nevertheless, there was a clear discrepancy in relation to the in situ data, measured between 1979 and 1982 at all stations, with a relative error of overestimation between 15.96 and 435 mm (Table 4). Another data discrepancy, though with lower relative error, occurred between 1995 and 1997, reaching 46.94 mm of underestimation.

Silva et al. (2012) studied monthly precipitation estimates derived from the gridded product of UDEL and the algorithm 3B43_V6, for Rio Grande do Norte, northeastern

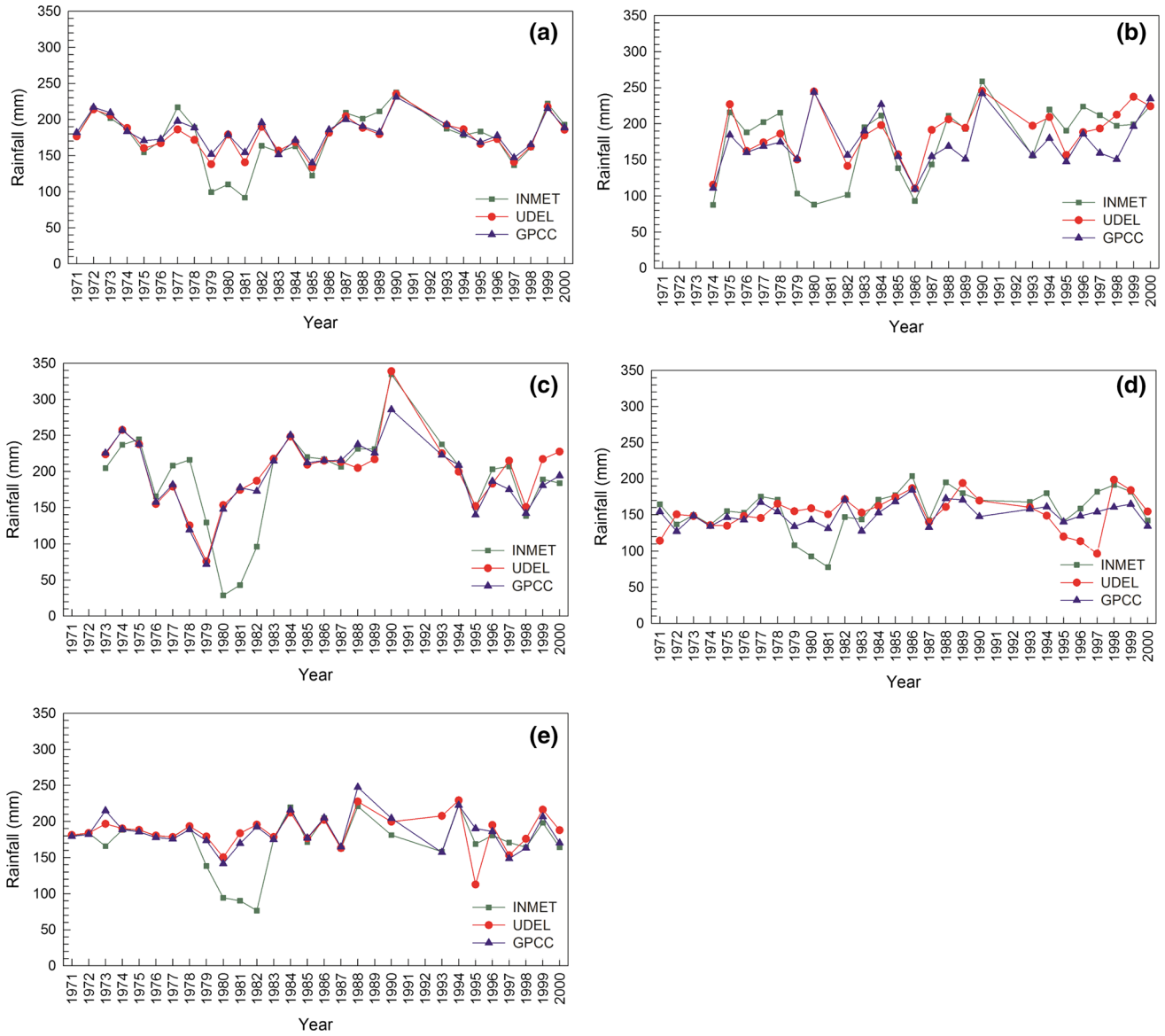


Fig. 2 Precipitation (mm) of interannual distribution for GPCC, UDEL and INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

Table 3 Correlation analysis between yearly rainfall estimated by the GPCC and the UDEL compared to those observed in INMET stations during historical series

Product	Cruzeiro do Sul	Eurinepé	Lábrea	Rio Branco	Tarauacá
	r	*r*	*r*	*r*	*r*
GPCC	0.91	0.79	0.83	0.95	0.87
UDEL	0.94	0.73	0.78	0.72	0.84

Brazil, and observed that the interannual variation in data behavior coincided with El Niño periods, when the rainfall pattern in the state is altered.

Several studies described the influence of climate variability by ENSO on changes in the precipitation regime in the Amazon region between 1979 and 1982 (Souza et al. 2000; Liebmann and Marengo 2001; Marengo 2006; Andreoli et al. 2012; Almeida et al. 2016). However, the Multivariate ENSO-MEI index (2014) indicated a long period of neutrality, with the occurrence of positive anomalies in the sea surface temperature (SST) only in late 1982, when the monthly precipitation products of GPCC and UDEL converged with the INMET data again.

In 1995 and 1996, the phenomenon of ENSO was also considered neutral, but in 1997 anomalies exceeded 2 °C, classifying this El Niño as strong. The same anomaly continued in 1998, although in this year the data of the precipitation product coincided with the INMET data again (Fig. 2). In other words, the cause of the discrepancy

Table 4 Relative error (mm) analysis on between rainfall estimated by the GPCC and the UDEL compared to those observed in INMET stations in selected years

Year	Product	Cruzeiro Sul	Eirunepé	Lábrea	Rio Branco	Tarauacá
1979	GPCC	53.08	46.42	–44.43	23.80	25.49
	UDEL	39.05	45.64	–41.43	43.29	29.62
1980	GPCC	62.54	177.47	415.61	53.65	50.33
	UDEL	62.69	178.60	435.53	70.97	59.62
1981	GPCC	68.36	–	313.22	68.32	88.11
	UDEL	53.29	–	305.86	93.63	103.66
1982	GPCC	19.68	54.42	79.59	16.43	151.67
	UDEL	15.96	39.58	94.67	16.76	155.58
1995	GPCC	–8.13	–22.63	–6.91	–6.46	12.83
	UDEL	–9.35	–17.77	1.18	–28.36	–33.17
1996	GPCC	2.87	–16.94	–8.18	–15.26	2.98
	UDEL	0.21	–15.87	–9.86	–46.94	7.96
1997	GPCC	7.64	–24.68	–15.54	–16.11	–13.06
	UDEL	3.27	–8.70	3.75	3.78	–10.33

between the data of the GPCC and UDEL products in the period analyzed was not related to the El Niño events.

In 1979, 1980, 1981 and 1997, when the dissimilarity between the observed and estimated data was greatest, an interhemispheric gradient between the northern and southern tropical Atlantic basins was observed, with positive anomalies in the northern basin (FUNCEME 2015). Several studies (Marengo et al. 2008, 2012 Zeng et al. 2008; Yoon and Zeng 2010) pointed out that this pattern was associated with negative rainfall anomalies in the Amazon, especially in the southern part.

This positive gradient was responsible for the low precipitation recorded at INMET stations. However, the data estimated by the gridded products did not follow this pattern, especially at the stations Eirunepé and Lábrea, where the relative errors were highest (Table 4) both for GPCC (177 mm for Eirunepé and 416 mm for Lábrea in 1980) and for UDEL (179 mm for Eirunepé and 436 mm for Lábrea in 1980).

Of the analyzed stations, Eirunepé is located farthest north and Lábrea farthest east (Fig. 1), which may explain the high relative error between the observed and estimated data for the products. This occurs because, due to the scarcity of data for the Amazon region, other stations located farther north and east of the region are considered in the composition of the data grid, which affects the result, since in these other areas the influence of positive anomalies of the North Atlantic SST pattern on the rainfall regime may be weaker than that of other phenomena.

Thus, the performance of other meteorological systems can help explain the discrepancy between the data for these specific periods. To this end, a more detailed study on the influence of mesoscale systems that can alter regional rain patterns is necessary. The reason is that, given the spatial resolution of the products reviewed here, they are unable to capture possible changes in regional rainfall patterns because the data underlying the grids are from a few scattered stations.

When observing the performance of products from a seasonal point of view (Fig. 3), observed and gridded precipitation had the same pattern, which again indicates the data agreement in the rainfall regime throughout the year. However, it was found that both products estimated rainfall rates best in the driest months, while in the rainiest months the discrepancy between the observed data by INMET and those estimated by grid products was greater. The exceptions were Rio Branco and Lábrea, where the performance of the UDEL product was unsatisfactory for both seasons.

Results similar to those of most stations were found by Sun et al. (2014) in a comparison with various gridded precipitation products for China (including those of GPCC and UDEL), who observed that these estimated rainfall better in the dry season.

The worse performance of estimates in the rainiest months can be explained by the higher occurrence of systematic measurement errors during rainfalls. The weather conditions (e.g., wind, air temperature and radiation) during a rainfall, as well as their intensity, affect the measurement quality due to evaporation (Schneider et al. 2014). In months with a lower number of rainfalls, errors of this nature are also minimized.

As the gridded products use data from several stations, which have systematic errors with different magnitudes, the dissimilarity of the estimates generated by these products tends to increase during the rainy season.

The agreement of GPCC (Fig. 4) and UDEL (Fig. 5) products for the data estimated throughout the climatological series was analyzed by regression analysis. This resulted in a model of the relationship between the

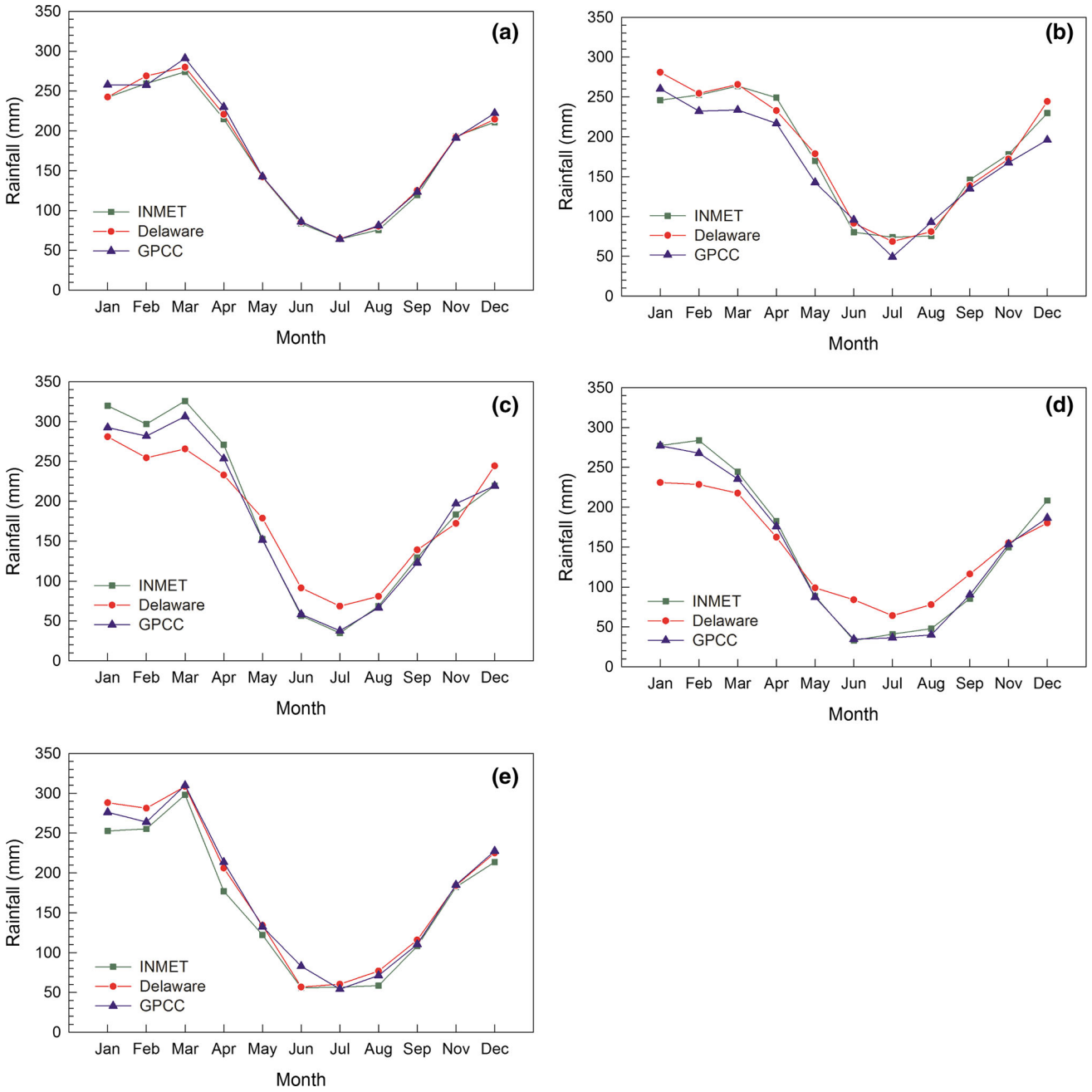


Fig. 3 Monthly distribution of average rainfall (mm) from the entire series for GPCC, UDEL and INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

observed rainfall values and those estimated by grid products.

Based on the regression coefficient r^2, it was found that the precipitation data of the station Cruzeiro do Sul were highly precise for the products of GPCC (0.82) and UDEL (0.89), whereas the precision of station Rio Branco was only high for the GPCC product (0.91). The precision at the other stations was satisfactory for both products.

According to Andrade-Júnior et al. (2003), the consideration of r^2 as the only definition criterion of quality of a product is not appropriate, because this coefficient cannot establish the type and magnitude of the differences. Therefore, the types of errors associated with the products must be taken into consideration.

Table 5 shows the RMSE, as well as its ratio in relation to the mean of the observed data. The lowest RMSE values for GPCC were observed at the stations Cruzeiro do Sul (44.9 mm) and Rio Branco (35.6 mm), and the highest precision for the same stations. For UDEL, the RMSE of station Cruzeiro do Sul was the best (35.4 mm).

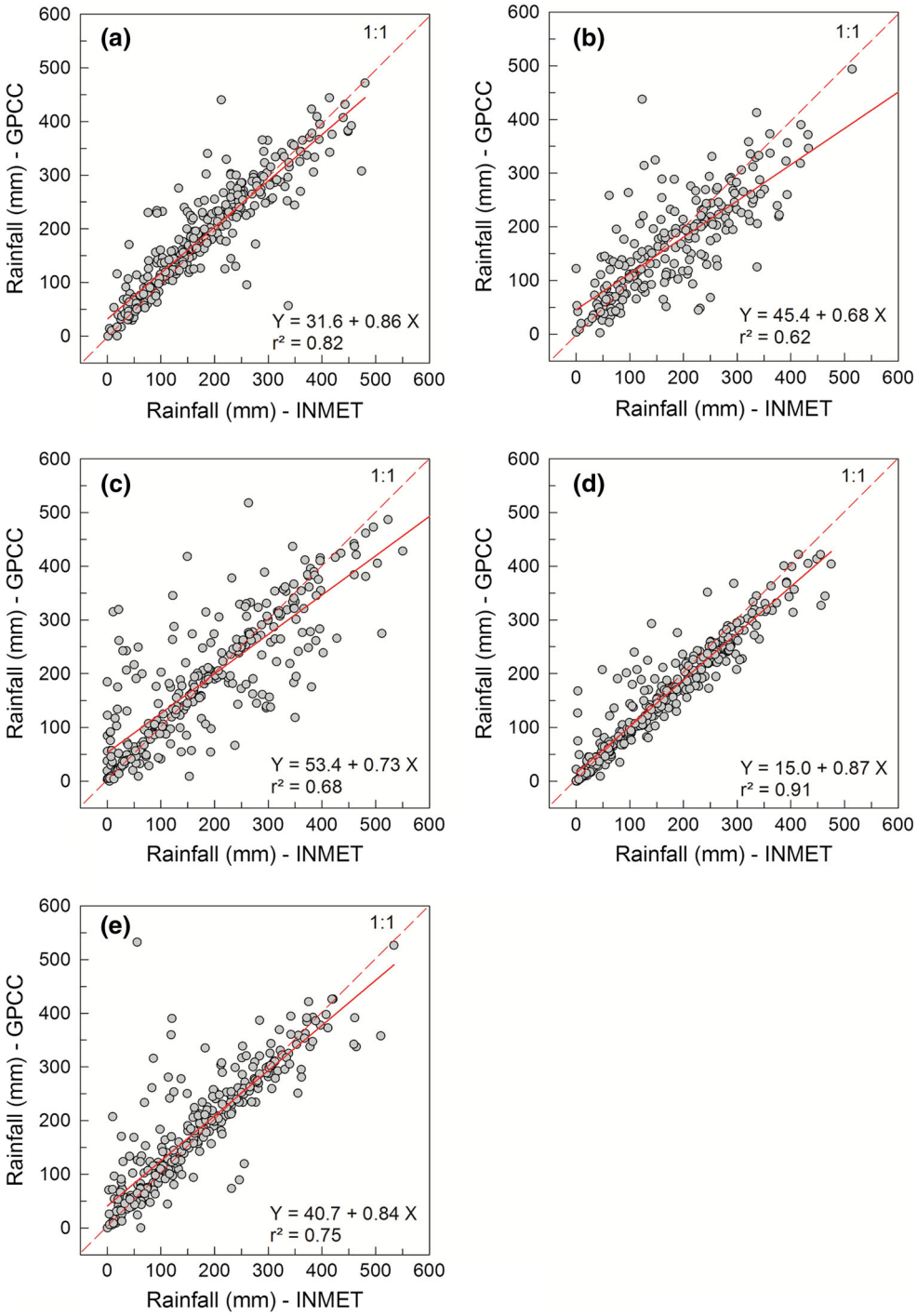


Fig. 4 Regression analysis between rainfall amounts estimated by the product of GPCC and values observed in INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

It is noteworthy that the error percentage of GPCC in relation to the precipitation mean over the time series was consistent with the ratio estimated by Rudolf et al. (1994) for four of the five stations. These authors analyzed data at a global level and estimated an error ratio between 7 and 40% in relation to the mean for regions with a low density of stations.

In general, it is understood that the error of good estimation methods is close to zero; however, the proportion of the error type is essential, i.e., whether it is random or systematic. For the precipitation products of GPCC and UDEL, more than 94% error (Table 6) with a non-systematic nature (MSE_u) was observed at all stations, which affects the precision of the estimates, but not necessarily their accuracy. The latter is mainly associated with the systematic errors related to estimation errors inherent in the GPCC and UDEL interpolation method itself, which were low for both products.

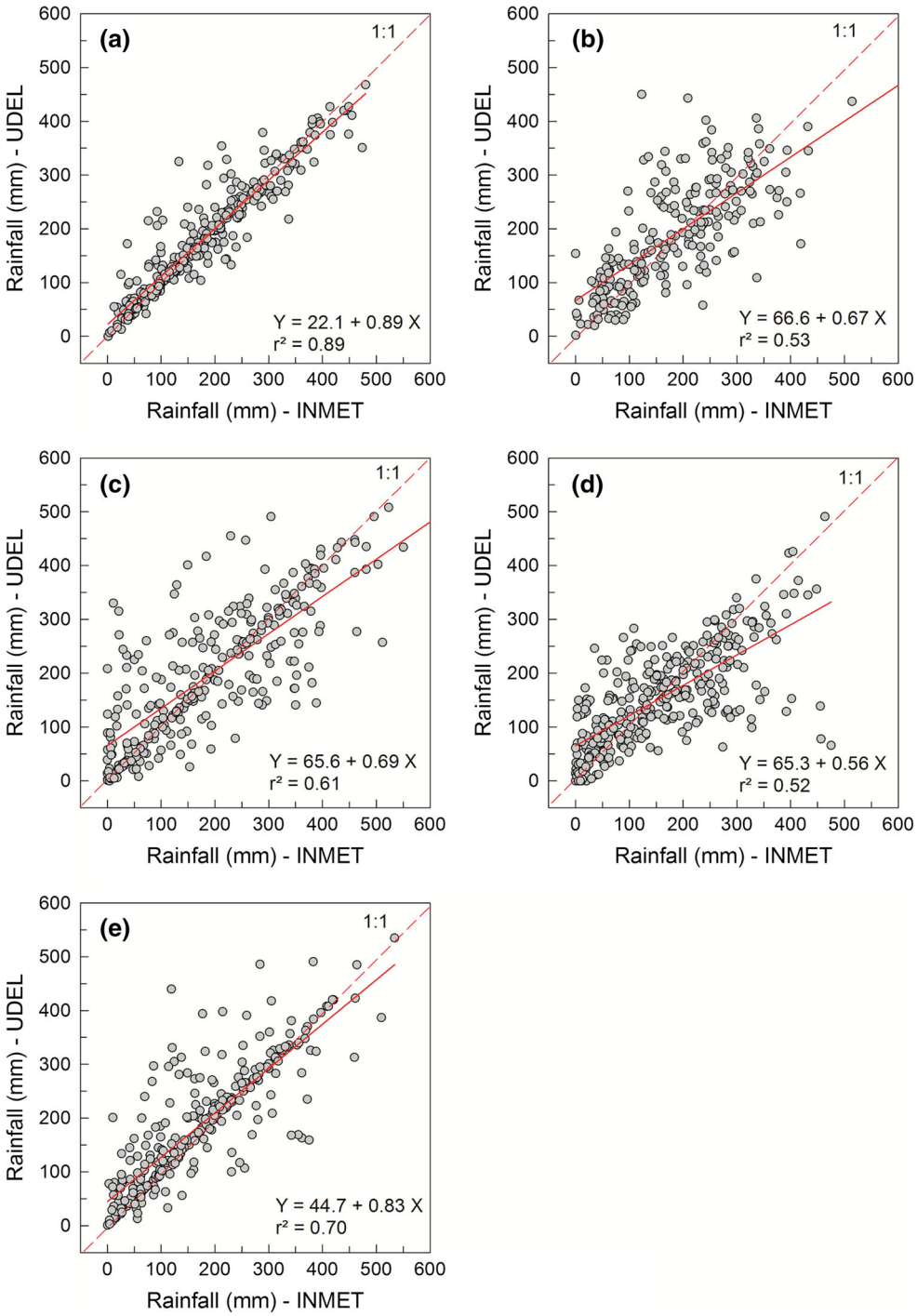


Fig. 5 Regression analysis between rainfall amounts estimated by the product of UDEL and values observed in INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

With respect to the accuracy of the Willmott coefficient (Table 7), the coefficient d of all stations was found to be higher than 0.82, mainly for Cruzeiro do Sul (0.95 in GPCC and 0.97 in UDEL) and Tarauacá (0.92 in GPCC and 0.91 in UDEL).

These results indicate a high accuracy level for the GPCC and UDEL products, i.e., a high degree of similarity between the estimated precipitation values and those observed in situ. However, the analysis of the results of the products showed that while the accuracy of four of the five stations analyzed was relatively similar, the station Rio Branco had a distinct pattern with regard to the products of GPCC and UDEL (coefficient *d* of 0.97 and 0.82, respectively). In the latter case, this value was the lowest of the five studied stations.

Finally, in terms of both accuracy and precision, the overall performance of the results was expressed by coefficient c (Table 7). The performances of the products of

GPCC and UDEL were found to be different for two stations. The performance of the station Rio Branco was classified as high for GPCC and poor for UDEL, while Eirunepé was rated good for GPCC and mean for UDEL.

Although both products proved accurate and moderately coincident with each other, the precipitation product of GPCC had a more homogenous overall performance for all stations.

To date, no studies comparing these products with each other were carried out for Amazonia. However, similar results were found by Sun et al. (2014) in a comparison of eight gridded products of precipitation and air temperature data from China, using weather station data as reference. The authors concluded that, although the products had a similar behavior as of 1950, the results of GPCC were better than those of UDEL for regions in China with low altitudinal gradient.

While UDEL interpolates the absolute precipitation data, the GPCC uses anomaly values to correct the sampling errors. According to Becker et al. (2013), this methodological difference may contribute to more accurate results in regions with low station density.

This result is consistent with findings of Chen et al. (2002), who compared four rainfall interpolation methods (Cressman, Barnes, Shepard, and the Optimal Interpolation—OI, of Gandin). It was found that estimates were more precise when the algorithms used anormalities instead of total rainfall values.

Table 5 Analysis of rainfall (mm) estimation errors generated by the GPCC and the UDEL from the mean of the monthly values observed in the INMET stations

Stations	INMET Mean (mm)	GPCC RMSE (mm)	GPCC % error	UDEL RMSE (mm)	UDEL % error
Cruzeiro Sul	174.6	44.9	25.7	35.4	20.2
Eirunepé	184.3	70.4	38.2	78.8	42.8
Lábrea	189.0	79.6	42.1	89.1	47.1
Rio Branco	156.8	35.6	22.7	79.3	50.5
Tarauacá	169.7	59.4	35.0	64.5	38.0

RMSE root-mean-square error

Comparison of products of monthly air temperature

An analysis of the interannual (Fig. 6) and monthly distribution of average temperature (°C) (Fig. 7) of the climatological series indicated that the products of GHCN and UDEL followed the tendency of air temperature

Table 6 Analysis of the types of errors estimation of rainfall (mm) generated by the GPCC and UDEL in relation to the observed monthly data of the INMET stations

Products	GPCC (Total Full V6)					UDEL (V3.01)				
Stations	Cruzeiro Sul	Eirunepé	Lábrea	Rio Branco	Tarauacá	Cruzeiro Sul	Eirunepé	Lábrea	Rio Branco	Tarauacá
$RMSE_s$	7.14	14.37	2.58	5.79	13.79	3.61	5.25	7.38	3.51	14.93
$RMSE_u$	44.35	68.89	79.53	35.15	57.77	35.18	78.67	88.82	79.19	62.72
$RMSE_a$	31.62	45.36	53.11	14.96	40.67	22.07	66.58	65.57	65.27	44.67
$RMSE_i$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
$RMSE_p$	34.48	59.73	50.53	20.75	26.88	18.46	61.33	58.19	68.79	29.74
MSE_s/MSE	0.025	0.042	0.001	0.026	0.054	0.01	0.004	0.007	0.002	0.054
MSE_u/MSE	0.975	0.958	0.999	0.974	0.946	0.99	0.996	0.993	0.998	0.946

$RMSE_s$ root-mean-square systematic error, $RMSE_u$ root-mean-square non-systematic error, $RMSE_a$ root-mean-square additive systematic error, $RMSE_i$ root-mean-square interdependent systematic error, $RMSE_p$ root-mean-square proportional systematic error, MSE_s mean systematic error, MSE_u mean non-systematic error, *MSE* total mean error

Table 7 Statistical performance assessment of the rainfall estimates generated by the GPCC and the UDEL with the data observed from the INMET stations

Stations	GPCC *d*	GPCC *c*	GPCC performance	UDEL *d*	UDEL *c*	UDEL performance
Cruzeiro Sul	0.95	0.86	Great	0.97	0.91	Great
Eirunepé	0.87	0.68	Good	0.84	0.61	Mean
Lábrea	0.88	0.72	Good	0.84	0.66	Good
Rio Branco	0.97	0.93	Great	0.82	0.59	Poor
Tarauacá	0.92	0.80	Very good	0.91	0.77	Very good

d Willmott index of agreement, *c* Camargo performance index

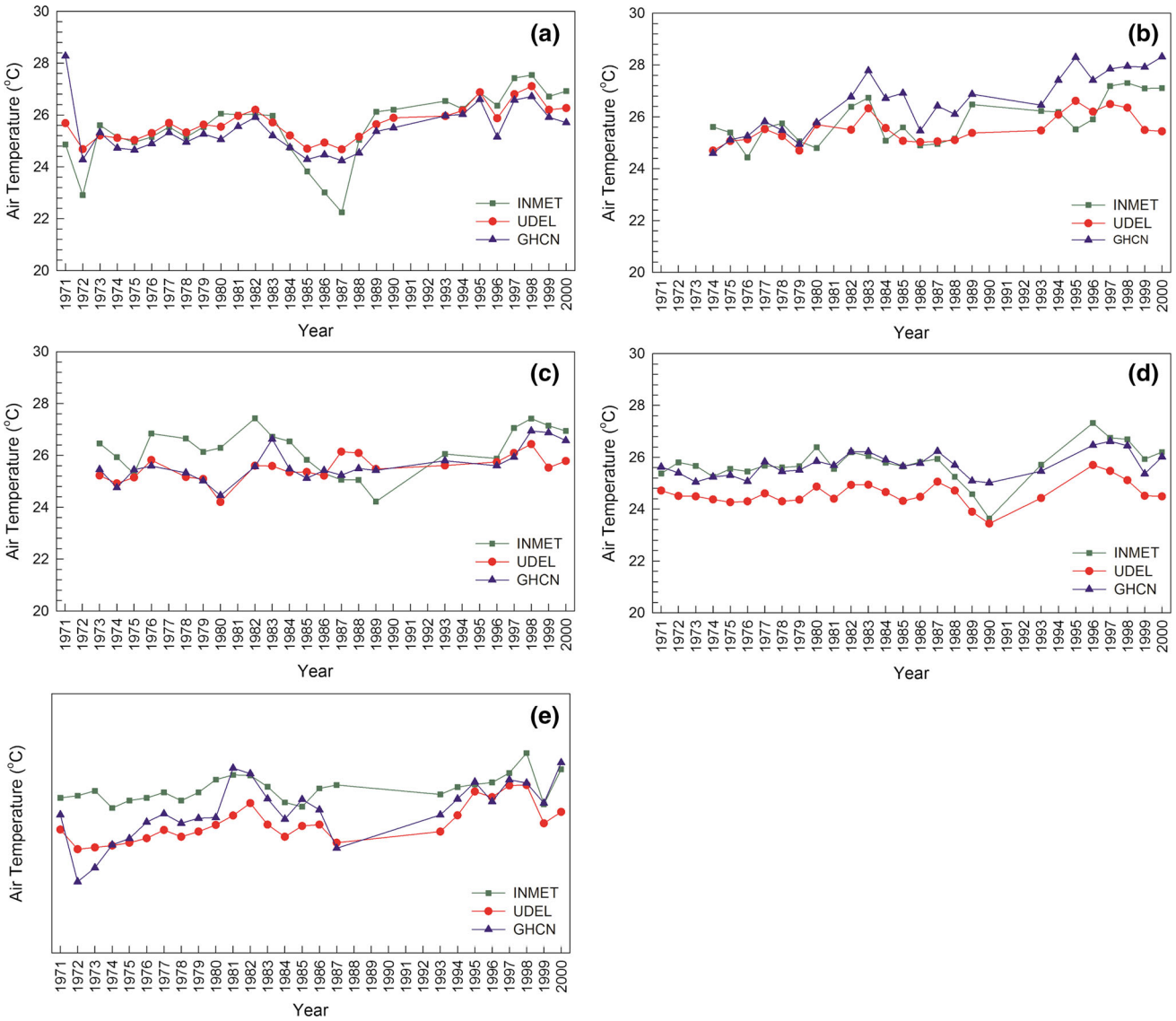


Fig. 6 Interannual temperature distribution for GHCN, UDEL and INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

observed at the INMET stations for most years, despite the discrepant absolute values, with median correlation coefficients (Table 8).

On the other hand, for the long-term variation of air temperature, the errors were reduced. Table 9 shows the RMSE as well as its ratio in relation to the mean of the observed data. A low RMSE value (≤1.6 °C) was observed for all stations and both products.

These values, although low, exceeded those of Willmott and Matsuura (1995), who obtained an error of only 0.38 °C when using the interpolator CAI with the DEM for air temperature data.

The difference in the uncertainty of estimates obtained by these authors and analyzed in our study can be explained, among other factors, by the fact that the number of weather stations in southwestern Amazonia is low.

The analysis of the error type (Table 10) highlighted that, in the GHCN product, the errors of data from all stations were mostly random (>98%), which affects the precision but not the accuracy of the estimates, as mentioned above. The random errors of data of UDEL were >82% for Cruzeiro do Sul, Eirunepé and Lábrea, while Rio Branco and Tarauacá had higher percentages of systematic errors (77 and 80%, respectively).

In the regression analysis between the observed values of mean air temperature and those estimated by the gridded products, it was found that the air temperature values for GHCN were moderately precise for Rio Branco

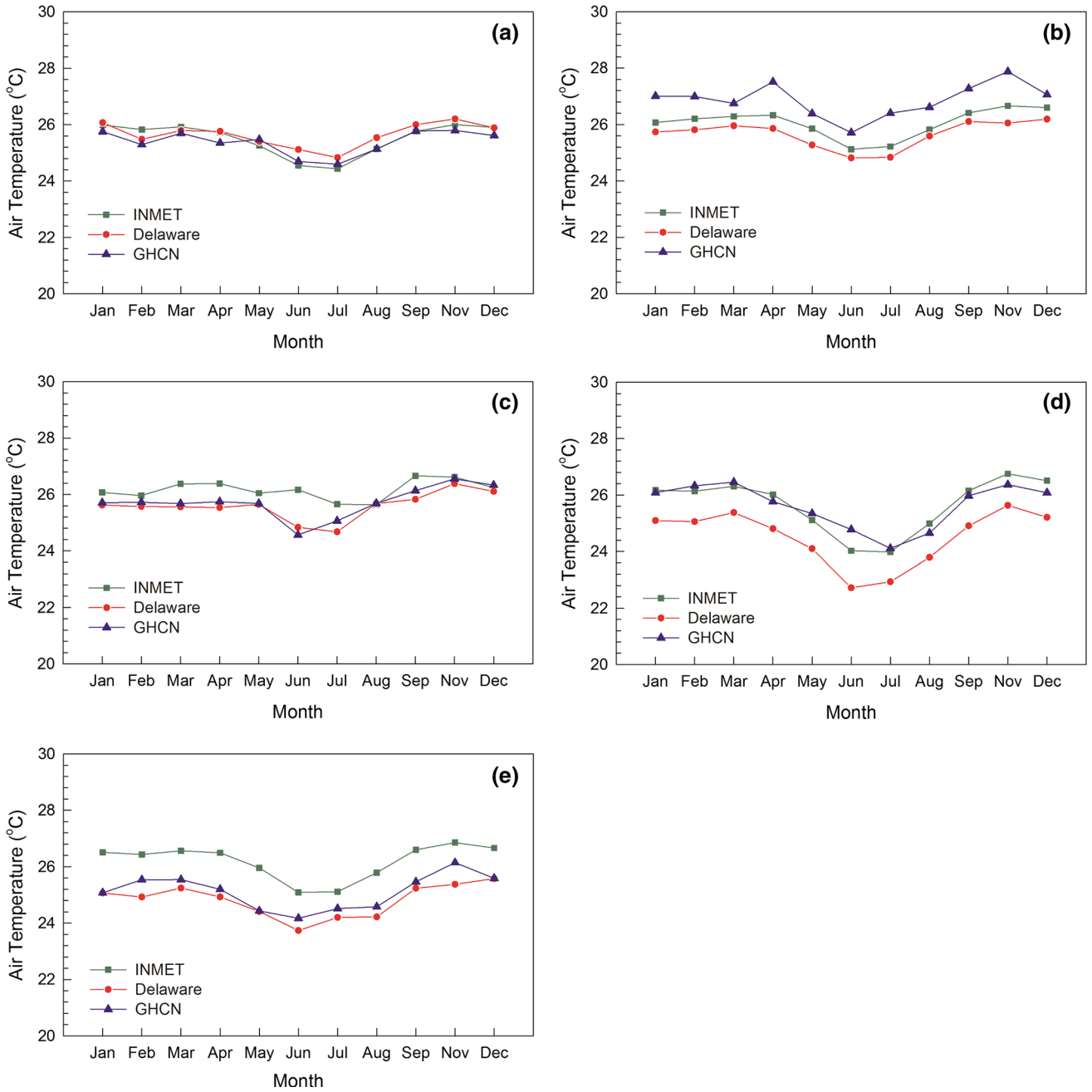


Fig. 7 Monthly distribution of average temperature (°C) from the entire series for GHCN, UDEL and INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

($r^2 = 0.61$) and little precise ($r^2 < 0.35$) for the other stations (Fig. 8).

The air temperature product of UDEL was more precise for all stations with low and/or medium precision, except for Rio Branco with $r^2 = 0.78$, indicating high precision for this product at this location (Fig. 9).

When considering the coefficient of Willmott (Table 11), similar accuracy values were found for the stations Eirunepé and Tarauacá. The other stations, however, had values that differed between GHCN and UDEL. These variations were mainly associated with the methodological differences used to generate these products.

The best agreement between UDEL (0.79) and GHCN (0.70) was obtained for the station Cruzeiro do Sul. This station is located at an altitude of 170 m asl and closest to the Andes of all analyzed stations. In an area with low density of meteorological observations, stations at higher altitudes can be used to interpolate air temperature values for the region of Cruzeiro do Sul. Since the interpolation

Table 8 Correlation analysis between the monthly air temperature estimated by GHCN and the UDEL compared to those observed in INMET stations during historical series

Product	Cruzeiro do Sul	Eurinepe	Lábrea	Rio Branco	Tarauacá
	r	*r*	*r*	*r*	*r*
GHCN	0.55	0.60	0.45	0.78	0.57
UDEL	0.72	0.60	0.18	0.89	0.73

Table 9 Analysis of monthly air temperature (°C) estimation errors generated by GHCN and the UDEL from the mean of the values observed in the INMET stations

Stations	INMET Mean (°C)	GHCN RMSE (°C)	GHCN % error	UDEL RMSE (°C)	UDEL % error
Cruzeiro Sul	25.5	1.3	5.2	1.1	4.2
Eirunepé	26.1	1.4	5.5	1.0	3.9
Lábrea	26.1	1.2	4.6	1.3	5.2
Rio Branco	25.7	0.8	3.0	1.3	5.0
Tarauacá	26.3	1.6	6.1	1.6	6.0

RMSE root-mean-square error

method used by UDEL is the CAI adjusted by DEM, this product tends to generate more accurate results than GHCN for locations closer to high altitude areas.

The accuracy for the stations Lábrea and Rio Branco was higher for GHCN, with d values of 0.59 and 0.88, respectively, while the Willmott coefficient was 0.25 and 0.75, respectively, for UDEL for these stations. Located in the east of the study area (Fig. 1), these stations lie in the so-called arc of deforestation of the Brazilian Amazon.

Several studies (Foley et al. 2003; Rocha 2001; Saad et al. 2010) indicate that the increase in deforestation provokes air temperature increases, due, among other factors, to the higher solar incidence, wind penetration and reduced soil moisture, leading to reduced ET. Thus, when analyzing the trend in air temperature over the time series, it is worth remembering that the land-use changes in the surroundings of stations hamper interannual comparisons (Ducre-Robitaille et al. 2003).

In this respect, the GHCN product of air temperature obtained more accurate results, possibly because the quality control of this product uses homogenization techniques of input data to remove the influence of land-use change on data from stations that are used for interpolation.

When considering accuracy as well as precision (Table 11), the air temperature product had similar performance for almost all stations. An exception was the station Cruzeiro do Sul [with c index 0.39 (GHCN) and 0.57 (UDEL)], with very poor and poor performance, respectively. Only the performance for Rio Branco was classified as good for both products. The performance of the other stations was classified as poor and/or very poor for GHCN as well as UDEL.

A comparison between the statistical indices of the estimated values of precipitation and air temperature showed that the performance of the latter was lower than of the former, especially when considering the overall performance by the c index and its classification, as proposed by Camargo and Sentelhas (1997).

However, this performance should be carefully analyzed, firstly because the limits of this classification are arbitrary (which is inherent to any classification proposal), and secondly because this index provides the same weight for precision and accuracy values. As these (rainfall and air temperature) are meteorological variables, the variation around the mean is naturally high, due to the occurrence of anomalous years, interfering with the precision of the results, but not necessarily with their accuracy.

Moreover, the proper systematic errors of the measurements of observations of CWS located in remote areas cannot be ignored. As the observed data are used as standard parameters for the analyses, possible associated errors

Table 10 Analysis of the temperature estimation errors (°C) generated by GHCN and UDEL with the data observed from the INMET stations

Products	GHCN (V2)					UDEL (V3.01)				
Stations	Cruzeiro Sul	Eirunepé	Lábrea	Rio Branco	Tarauacá	Cruzeiro Sul	Eirunepé	Lábrea	Rio Branco	Tarauacá
$RMSE_s$	0.123	0.826	0.439	0.035	2.118	0.136	0.407	0.561	1.148	1.386
$RMSE_u$	2.226	11.092	6.114	0.573	15.727	1.052	0.923	1.226	0.558	0.746
$RMSE_a$	14.006	7.584	14.125	6.342	1.593	13.861	14.597	22.141	1.883	0.908
$RMSE_i$	0.00	0.00	0.00	0.00	1.294	0.00	0.00	0.00	0.00	0.932
$RMSE_p$	14.129	6.758	14.564	6.377	0.525	13.724	15.005	22.702	3.031	0.479
MSE_s/MSE	0.003	0.006	0.005	0.004	0.018	0.016	0.163	0.173	0.809	0.775
MSE_u/MSE	0.997	0.994	0.995	0.996	0.982	0.984	0.837	0.827	0.191	0.225

$RMSE_s$ root-mean-square systematic error, $RMSE_u$ root-mean-square non-systematic error, $RMSE_a$ root–mean-square additive systematic error, $RMSE_i$ root-mean-square interdependent systematic error, $RMSE_p$ root-mean-square proportional systematic error, MSE_s mean systematic error, MSE_u mean non-systematic error, *MSE* total mean error

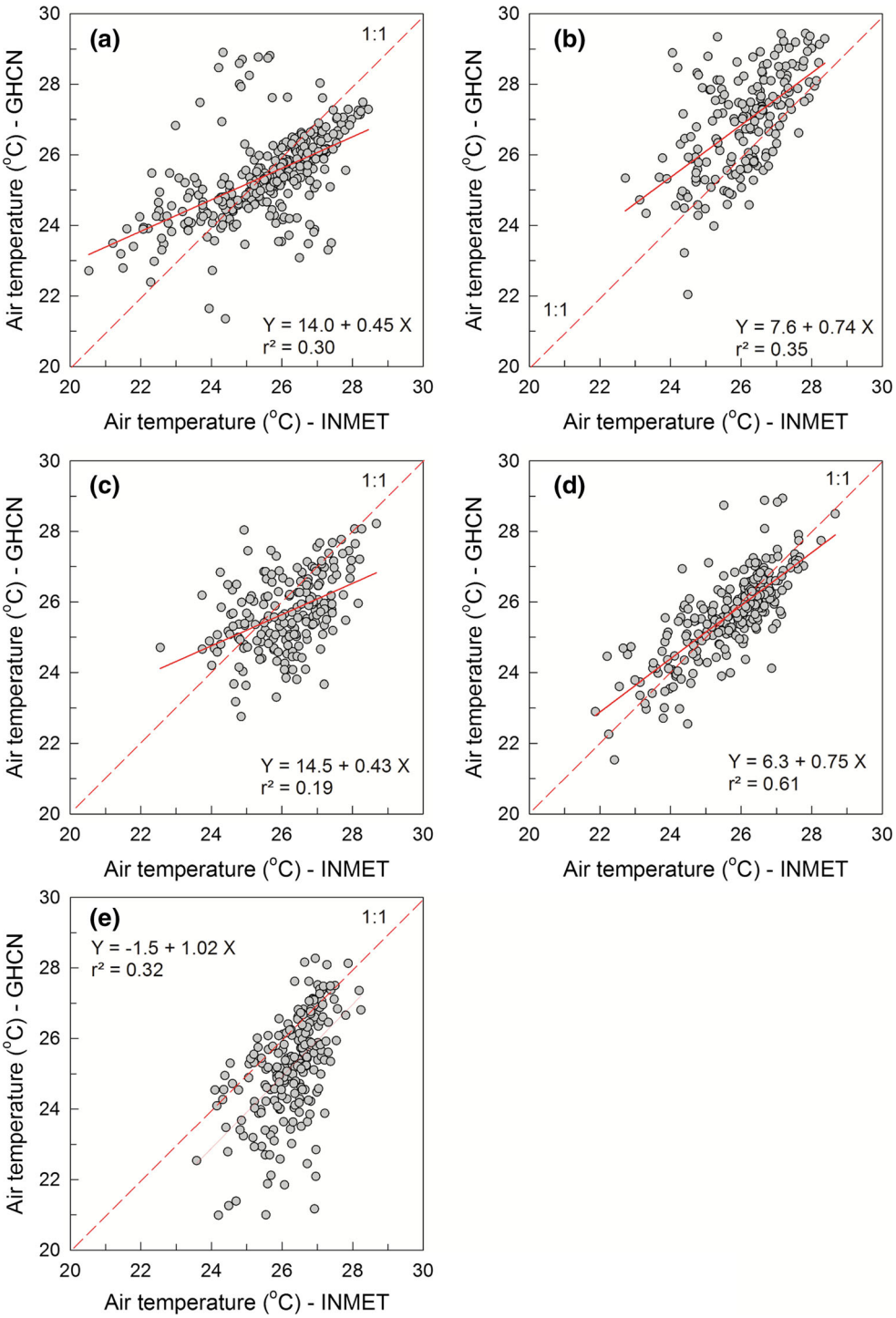


Fig. 8 Regression analysis between the temperature values estimated by the product of GHCN and values observed in INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

compromise the quality of the comparisons (Schneider et al. 2014). Thus, both issues can generate outliers, which influence the regression analysis and results of statistical indices and consequently the precision and accuracy of the estimates.

Therefore, in regions where no long-term weather data series with adequate spatial distribution are available, the existence of estimated air temperature values reasonably close to the observed can be considered satisfactory.

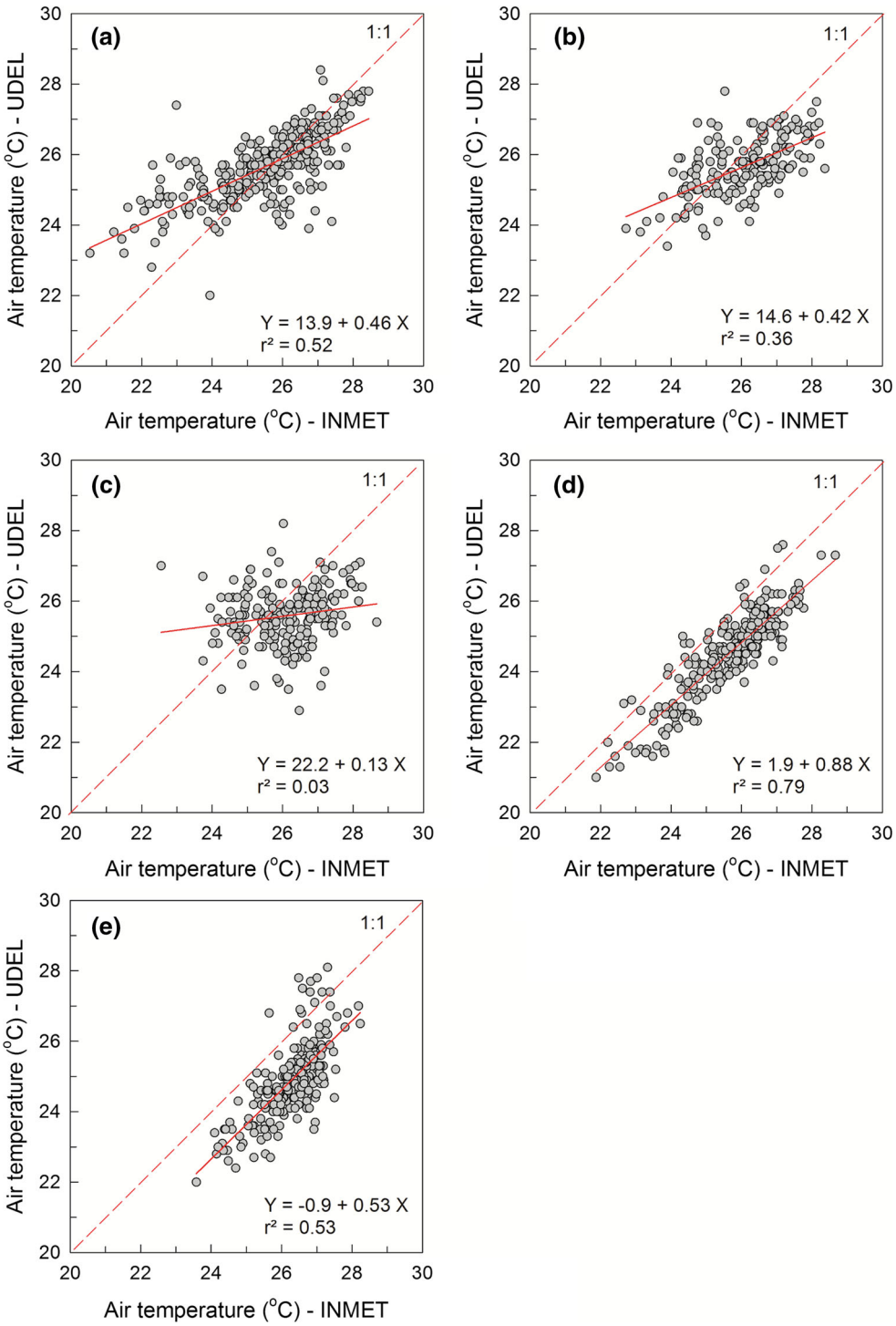


Fig. 9 Regression analysis between the temperature values estimated by the product of UDEL and values observed in INMET stations: **a** Cruzeiro do Sul; **b** Eirunepé; **c** Lábrea; **d** Rio Branco; **e** Tarauacá

Conclusions

The different techniques of data quality control and interpolation methods indicated minor differences between precision and accuracy of the gridded products studied. With regard to precipitation, the results suggest that the products of GPCC as well as of UDEL adequately represent the precipitation variability over years, although for GPCC, the differences between the stations were smaller.

With respect to the air temperature behavior, the precision of GHCN and UDEL was low, while the accuracy was moderate, although the differences between products were significant for some stations.

Table 11 Statistical performance assessment of the air temperature estimate generated by GHCN and UDEL with the data observed from the INMET stations

Stations	GHCN *d*	GHCN *c*	GHCN performance	UDEL *d*	UDEL *c*	UDEL performance
Cruzeiro Sul	0.70	0.39	Very bad	0.79	0.57	Poor
Eirunepé	0.65	0.39	Very bad	0.68	0.41	Bad
Lábrea	0.59	0.27	Very bad	0.25	0.04	Very bad
Rio Branco	0.88	0.69	Good	0.75	0.66	Good
Tarauacá	0.52	0.30	Very bad	0.53	0.38	Very bad

d Willmott index of agreement, *c* Camargo performance index

In view of the lack of long-term data for the south-western Amazon, depending on the study target, the gridded temperature products discussed here can be used in climate research, provided the uncertainty degree is taken into account in the analysis of results.

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